FinVerify Optimization

SQL Analysis Task 1 of 3

Onboarding Funnel Conversion

Measures how many unique users complete each stage of the onboarding journey in April 2026, and calculates the drop-off rate between stages.

SECTION 01: SQL Query

```
WITH funnel AS ( SELECT COUNT(DISTINCT user_id) AS total_users, COUNT(DISTINCT CASE WHEN
stage_name = 'phone_verified' THEN user_id END) AS phone_verified, COUNT(DISTINCT CASE WHEN
stage_name = 'profile_completed' THEN user_id END) AS profile_completed, COUNT(DISTINCT CASE
WHEN stage_name = 'kyc_submitted' THEN user_id END) AS kyc_submitted, COUNT(DISTINCT CASE WHEN
stage_name = 'account_activated' THEN user_id END) AS account_activated FROM
user_onboarding_events WHERE event_timestamp >= '2026-04-01' AND event_timestamp <
'2026-05-01' ) SELECT total_users, phone_verified, profile_completed, ROUND(profile_completed
* 100.0 / NULLIF(phone_verified, 0), 2) AS profile_completion_rate, kyc_submitted,
ROUND(kyc_submitted * 100.0 / NULLIF(profile_completed, 0), 2) AS kyc_submission_rate,
account_activated, ROUND(account_activated * 100.0 / NULLIF(kyc_submitted, 0), 2) AS
activation_rate FROM funnel;
```

SECTION 02: Expected Output

Actual output values depend on the underlying dataset. The table below illustrates the expected output format:

total_users	phone_verified	profile_completed	profile_completion_rate	kyc_submitted	kyc_submission_rate	account_activated	activation_rate
1000	900	810	90.00	720	88.89	680	94.44

SECTION 03: Query Explanation

Step 1 — CTE (WITH funnel AS …): We first create a temporary result set called funnel. Inside it, we scan the user_onboarding_events table once and count how many unique users reached each stage. Using COUNT(DISTINCT user_id) inside a CASE WHEN statement ensures we only count a user once per stage, even if they triggered the same event multiple times.

Step 2 — Date Filter: The WHERE clause restricts data to April 2026 only (event_timestamp >= '2026-04-01' AND < '2026-05-01'). Using < instead of <= on the end date avoids accidentally including the first second of May.

Step 3 — Conversion Rates: In the outer SELECT, we calculate percentage conversion between consecutive stages. For example, profile_completion_rate = (profile_completed / phone_verified) × 100. We wrap the denominator in NULLIF(…, 0) so the query never divides by zero.

Step 4 — ROUND: All rates are rounded to 2 decimal places using ROUND(…, 2) for clean, readable output suitable for dashboards and reports.

SECTION 04: Business Impact

01 Identify Drop-off Hotspots: See exactly which stage loses the most users — e.g. if only 70% move from profile to KYC, that stage needs UX attention.

02 Prioritise Engineering Resources: Low conversion at KYC submission could indicate a buggy flow; the data provides evidence to escalate a fix.

03 Month-over-Month Benchmarking: Running this query each month creates a time-series of funnel health, letting the product team track whether changes improve conversion.

04 Acquisition ROI: Combining this with acquisition_channel (from the users table) reveals which channels bring users who actually complete onboarding.